Analytic Fourier Spectra of Dyadic and Growth Subsequences in the Syracuse Problem

Chen Mao
Contact: savy.ch@gmail.com, 1003838980@qq.com

Abstract: We present two families of analytic closed-form Fourier spectra for Collatz-associated sequences. First, for the v2(3n+1) sequence, we derive a fractal-exponent DFT closed form, proving its spectrum consists purely of dyadic lattice lines (a zero-difficulty baseline). Second, for the growth phase $n=2^K-1$, we prove its log-height spectrum is a strict Nyquist single line, with a provable extension via LTE. Additionally, the v2 balance lemma rigorously establishes the 4^{-1} inverse law for wavelet details. These results serve as calibrated "fossils" for excluding pseudo-difficulties in spectral analysis.

1. Dyadic Closed-Form Spectrum of v2 (S-E2)

Theorem 2 (v2 Fractal Spectrum) Let $N=2^K$ and define *a[n]=v2(3(2n+1)+1)*. Its DFT is given exactly by:

$$G[k] = \Sigma\_\{i=1\}\hat{}\{K\}\ \mathbb{1}[(N/2^i) \mid k] \cdot (N/2^i) \cdot \omega^{k\cdot e_i} + \omega^{k\cdot e_{base}}$$

with fractal exponents $e_i = (2\cdot 4^{i-1} - 2)/3$, $e_{base} = (2^{2K+1}-2)/3$. This closed form is verified to numerical precision (max_diff=1.6e−14 for N=256) and predicts amplitude ratios G[N/2]:G[N/4] = 100:50. This spectrum contains no Collatz difficulty and serves as the zero-baseline for residual comparisons.

2. Growth-Phase Pure Nyquist Theorem (S-G1~G3)

Theorem 3 (Growth Baseline) For $n=2^K-1$, the first 2K iterates obey the closed forms:

$$T^{2m}(2^K-1) = 3^m\cdot 2^{K-m} - 1,$$
$$T^{2m+1}(2^K-1) = 3^{m+1}\cdot 2^{K-m} - 2$$

Consequently, the detrended log-height spectrum is a **pure Nyquist single line** (second harmonic = 0%), with exact slope $(\log_2 3-1)/2$. For even K, LTE ($v_2(3^K-1)=2+v_2(K)$) extends the provable segment to $2K+2+v_2(K)$ steps, bypassing the sampling bottleneck that typically relies on the Collatz conjecture for long orbits.

3. v_2 Balance Lemma and Wavelet Mechanism (S-L3)

Theorem 4 (Balance Lemma) On any block of 2^m consecutive odd indices:

$$\Sigma\ (v_2(3(2j+1)+1) - 2) = v_2(3j_0+2) - m - 1$$

This sum is bounded and decreases monotonically with m (trivial scaling). This lemma rigorously proves the 4^{-l} inverse law for wavelet detail energy E(l) (S-J1), elevating a prior numerical observation to a deterministic structural theorem.

Keywords: Analytic spectrum · v_2 fractal · growth baseline · LTE extension · wavelet mechanism